Jyothi Rajan Ambika Kairali

AI & MACHINE LEARNING ENGINEER | AI AUTOMATION & AGENTIC RAG

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Portfolio: https://jyothi-ai-portfolio.vercel.app/ | GitHub: https://github.com/Jyothi162

PROFESSIONAL SUMMARY

AI & Machine Learning Engineer with 7+ years in data engineering, ETL, and automation, now building agentic AI and LLM applications while completing an M.Sc. in Artificial Intelligence at IU International University of Applied Sciences, Berlin. Built secure, containerized Retrieval-Augmented Generation (RAG) assistants and automated LLM workflows using Python, LangGraph, LangChain, Qdrant, Ollama, FastAPI, and Docker, with JWT authentication, role-based access control, hybrid vector search, and automated testing (21 passing pytest tests). Fluent in English; German B1 in progress (VHS Berlin).

TECHNICAL SKILLS

AI & LLM Engineering: Retrieval-Augmented Generation (RAG), AI agents, LangGraph, LangChain, tool calling, prompt engineering, embeddings, semantic and hybrid search, BM25, reciprocal rank fusion, RAG evaluation

Programming & ML: Python, SQL, PL/SQL, Unix shell scripting, PyTorch, Hugging Face Transformers, DistilBERT, scikit-learn, pandas, NumPy, NLP, classification evaluation

Backend, Security & Testing: FastAPI, REST APIs, JWT, role-based access control (RBAC), document-level authorization, pytest, prompt-injection testing, human-in-the-loop approval

Data, Deployment & Tools: Qdrant, ChromaDB, SQLite, Oracle, ETL pipelines, Selenium, Docker, Docker Compose, Ollama, Streamlit, Git/GitHub, Linux, Microsoft Dynamics CRM

SELECTED AI PROJECTS

Enterprise Agentic RAG Assistant | *Python | LangGraph | LangChain | Qdrant | Ollama | FastAPI | Docker | pytest*

- Developed an enterprise knowledge assistant with locally hosted Llama 3.2, vector retrieval, FastAPI endpoints, and a Streamlit interface.
- Implemented JWT authentication, document-level access controls for Employee, HR, IT, and Admin roles, and restricted TXT/PDF/DOCX ingestion.
- Containerized services with Docker Compose; validated authentication, authorization, and upload security with 21 passing pytest tests.

GitHub: https://github.com/Jyothi162/enterprise-agentic-rag

EnergyOps AI — Agentic Support Automation | *Python | LangGraph | ChromaDB | BM25 | SQLite | FastAPI | Docker*

- Built an agentic solar-energy support assistant combining local LLM inference, RAG, customer-request classification, and automated workflows.
- Implemented hybrid retrieval with BM25 and reciprocal rank fusion; achieved 100% Source Hit Rate@2 on a five-question evaluation set.
- Designed installation-status lookup, safety escalation, and human-approved ticket creation; tested routing, retrieval, and baseline prompt-injection scenarios.

GitHub: https://github.com/Jyothi162/energyops-ai

DistilBERT Sentiment Classification | *Python | Hugging Face | PyTorch | scikit-learn*

- Fine-tuned DistilBERT for Amazon product-review sentiment classification; achieved 90.75% accuracy and 90.70% F1-score on held-out validation data.

GitHub: https://github.com/Jyothi162/DistilBERT-Sentiment-Classification

PROFESSIONAL EXPERIENCE

Data Engineer — Stecktra Technologies Pvt. Ltd. (Client: Yodify Inc.) | *India | Aug 2020 – Jan 2024*

- Extracted product-catalog data from multiple websites using Python and Selenium; cleaned and structured records for database integration.
- Developed and optimized ETL workflows using Oracle PL/SQL and Unix shell scripting, automating recurring data-processing tasks.
- Loaded and maintained structured product data in Oracle databases, supporting consistent downstream data use.

Data Analyst — Stecktra Technologies Pvt. Ltd. (Client: Telematic Controls Inc.) | *India | Jan 2016 – Jan 2020*

- Processed product catalogs, specifications, and variations according to client requirements; prepared structured data for Microsoft Dynamics CRM.
- Validated data accuracy and consistency, supported development and testing assignments, and collaborated with cross-functional teams.

EDUCATION

M.Sc. Artificial Intelligence — IU International University of Applied Sciences | *Berlin | Apr 2025 – Present*

B.Tech, Electronics and Communication Engineering — The Rajaas Engineering College, Anna University | *India | Sep 2008 – Jul 2012*

ADDITIONAL PROJECTS

TransGo Weather Assistant: Rasa NLU chatbot with OpenWeatherMap API, intent/entity recognition, and a web-based chat interface. GitHub: https://github.com/Jyothi162/my_Rasa_project

Ideal Function Selection & Data Mapping: Python, pandas, NumPy, SQLite, MSE-based function selection, visualization, and unit tests. GitHub: https://github.com/Jyothi162/Ideal-Function-Selection-and-Data-Mapping

LANGUAGES

English: Fluent | German: B1 (in progress) | Hindi